



AI-Based Driver Monitoring System for Drowsiness and Distraction Detection (DriveGuard)



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Overview

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Background of the Problem

- Major contributor to road accidents
- Reduced alertness and reaction time due to fatigue
- Driver distraction (mobile usage, off-road gaze)
- Lack of continuous driver state monitoring in existing systems



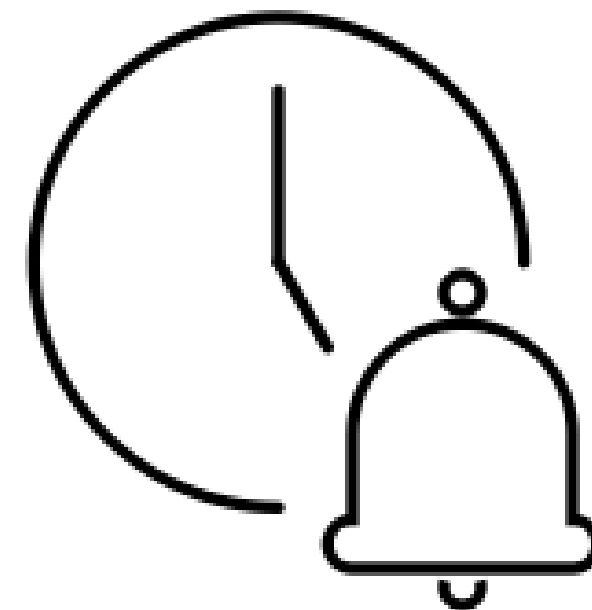


Rationale and Significance

- Accident prevention through early detection
 - Non-intrusive, camera-based monitoring
 - Cost-effective and practical for real vehicles
 - Supports intelligent vehicle safety systems
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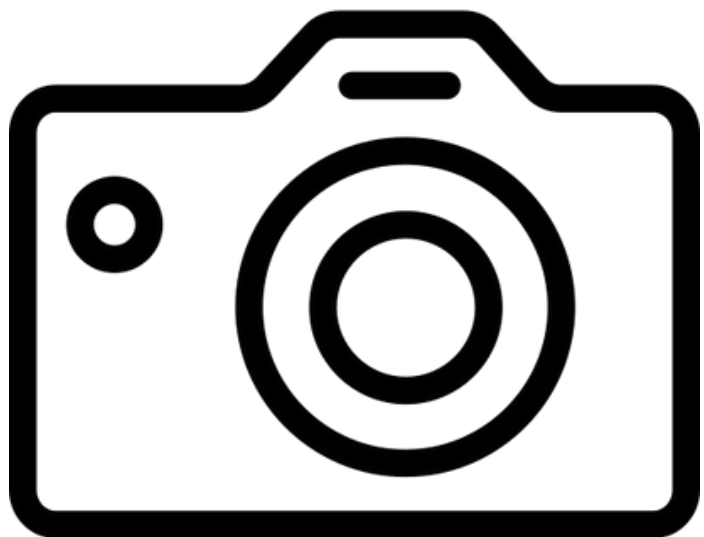
Objectives

- Detect driver drowsiness using facial cues
- Detect driver distraction (gaze, mobile usage)
- Generate real-time alerts for unsafe driver states
- Evaluate system performance and robustness

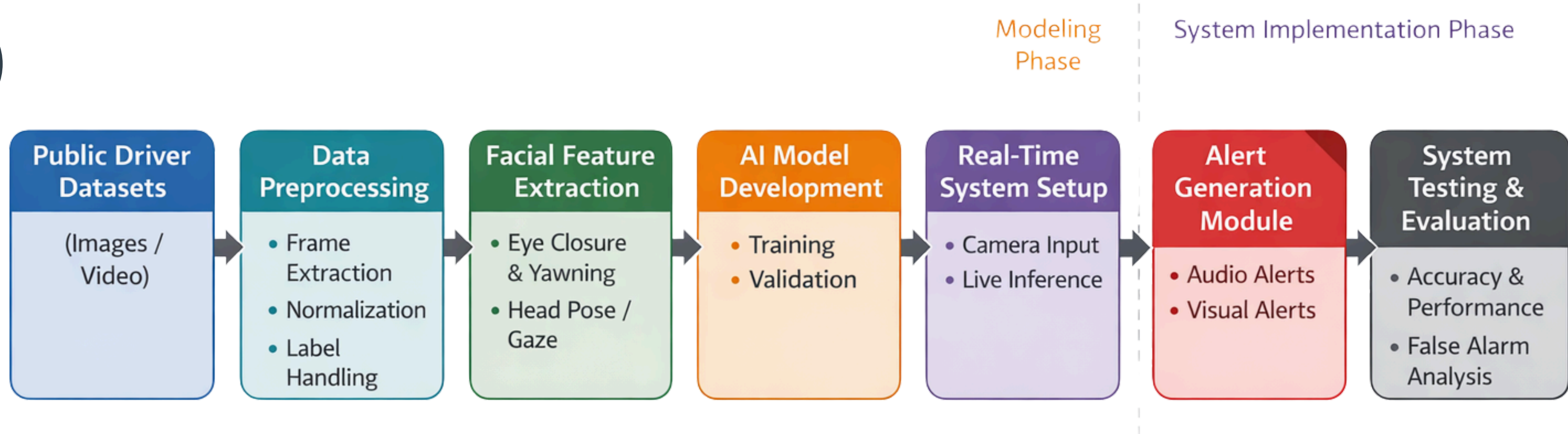


Proposed Methodology

- Camera-based driver video acquisition
- Facial cue extraction (eye closure, yawning, head pose)
- AI-based classification of driver state
- Real-time audio / visual alert generation



Proposed Methodology



Proposed Software and Hardware

Software

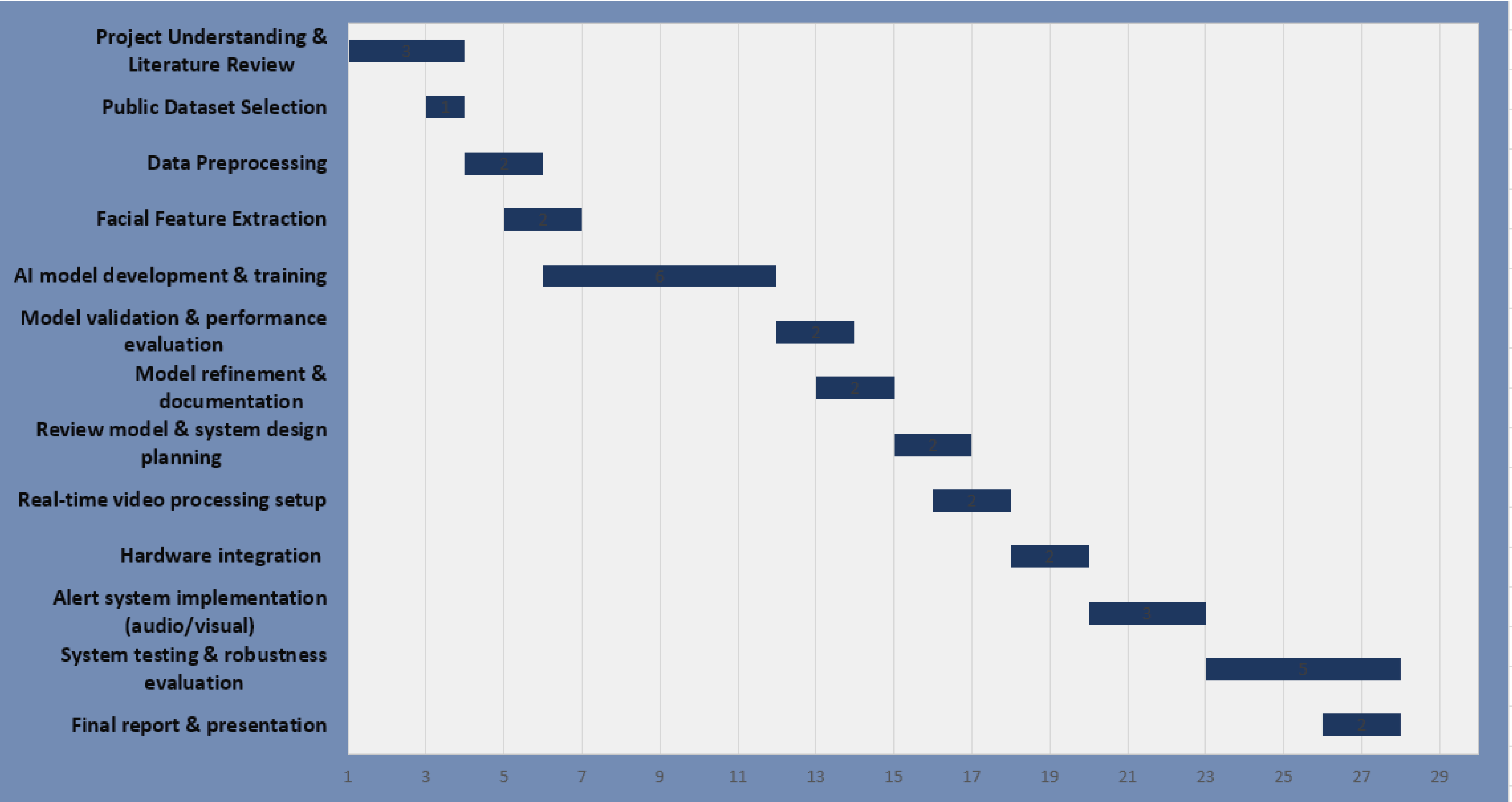
- Python
- OpenCV (video processing)
- MediaPipe / Dlib (facial feature extraction)
- TensorFlow / PyTorch (AI model development)

Hardware

- Laptop / PC
- Built-in or external webcam
- Speaker or buzzer for alerts



Project Timeline





Conclusion

- Proposed an AI-based driver monitoring system to detect drowsiness and distraction
 - Camera-based, non-intrusive approach suitable for real-world vehicles
 - Clear methodology with software-focused development followed by system integration
 - Realistic timeline ensuring feasibility within the project duration
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Thank You!

Q & A Session

